



Kurinda: A Machine Learning Early-Warning System for Predicting Village-Level Chronic Childhood Stunting Risk in Rwanda Using Multi-Source Data Fusion

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List of Acronyms/Abbreviations

Acronym	Meaning
ALU	African Leadership University
AI	Artificial Intelligence
API	Application Programming Interface
AUC-ROC	Area Under the Receiver Operating Characteristic Curve
CHIRPS	Climate Hazards Group InfraRed Precipitation with Station Data
CHW	Community Health Worker
DHIS2	District Health Information System 2
DHS	Demographic and Health Survey
EICV	Integrated Household Living Conditions Survey (Enquête Intégrale sur les Conditions de Vie des ménages)
ERD	Entity Relationship Diagram
FBF	Fortified Blended Foods
FEWS NET	Famine Early Warning Systems Network
GDP	Gross Domestic Product
HungerMap LIVE	World Food Programme's near real-time global hunger monitoring platform

IPC	Integrated Phase Classification
LightGBM	Light Gradient Boosting Machine
LIME	Local Interpretable Model-agnostic Explanations
ML	Machine Learning
MLOps	Machine Learning Operations
MoH	Ministry of Health
MODIS	Moderate Resolution Imaging Spectroradiometer
NDVI	Normalized Difference Vegetation Index
NECDP	National Early Childhood Development Program
NISR	National Institute of Statistics of Rwanda
OSM	Open Street Map
RAB	Rwanda Agriculture and Animal Resources Development Board
RAG	Retrieval-Augmented Generation
SDG	Sustainable Development Goal
SHAP	Shapley Additive Explanations
SMART	Specific, Measurable, Achievable, Relevant, Time-bound

SMOTE	Synthetic Minority Over-sampling Technique
SMS	Short Message Service
SPRP	Stunting Prevention and Reduction Project
UML	Unified Modeling Language
UNICEF	United Nations Children's Fund
WASH	Water, Sanitation and Hygiene
WHO	World Health Organization

ABSTRACT

Childhood stunting has been one of the most well-known public health problems in Rwanda; the Rwanda Demographic and Health Survey (2024-2025) found that about 510,000 children (or 27%) under the age of five in Rwanda are chronically undernourished (NISR, MoH & ICF, 2025). Although stunting is down from 33% in 2020, existing interventions are reactive: children are identified as stunted after irreversible brain and body damage is done in the first 1,000 days from birth. The existing predictive systems focus on AC malnutrition and neither operate at the village level where Rwanda's 45,000 CHWs work, nor are any studies conducted where machine learning might predict the progression of AC to CM. Existing machine learning studies predict AC malnutrition, and all to date were either retrospective or cross-sectional; they did not make predictions regarding the progression from AC to CM, and they were country-level rather than village-level (e.g., Rwanda has 45,000 CHWs). Kurinda (Kinyarwanda for “to protect”), this work proposes to contribute to developing an early-warning system using machine learning algorithms to forecast chronic stunting risk at the village level three to six months ahead of time, taking advantage of nutrition indicators from the DHS, data on crop production from the Rwanda Agriculture Board, food prices from eSoko, and satellite-derived climate signals (CHIRPS rainfall and MODIS NDVI). The model is built using the gradient-boosted trees (LightGBM) model with explainability using SHAP and delivered to users through the Next.js dashboard, a Kinyarwanda chatbot, and Africa's Talking SMS alerts. The system aims for a minimum of 70% AU-CR G at a 3-month forecasting time. The assumption underlying this study is that operationally actionable predictions of stunting risk at the village level, before stunting becomes clinically apparent, can be generated from multiple sources of data to relieve health care resources from being used reactively rather than preventively.

CHAPTER ONE: INTRODUCTION

1.1 Introduction and Background

Worldwide, stunted children (short stature relative to age, z score > 2 SD below the median of the WHO Child Growth Standards) are a public health emergency. In 2024, over 150 million children under the age of five were stunted worldwide (UNICEF, WHO & World Bank, 2024), about one-third of which was in sub-Saharan Africa. Stunting is a marker of chronic undernutrition during the first 1,000 days of life and affects mental function, immune system, lifetime income, and country economic productivity in an irreversible way (World Bank, 2024).

Despite progress over the past decade, Rwanda is a country facing a big challenge of undernutrition among children. The rate of stunting among children under 5 in the Rwanda Demographic and Health Survey (Rwanda DHS 2024) is 27%, whereas it was 33% in the Rwanda DHS 2020. It is estimated that 25 % of under-5 children are victims of stunted growth, and in rural Rwanda, the cases are over 510,000, with children between 18 and 23 months having rates exceeding 36%. While Rwanda's current rate of decline will not allow the Vision 2050 goals of reducing stunting cases to 5.5% in 2035 and 3% in 2050, a more radical shift in the way of targeting interventions will increase the likelihood of achieving these objectives.

Traditional interventions for the prevention and/or treatment of stunting are programmatic and reactive, addressing stunting once a child has been screened and identified as being at risk by routine CHW visits (UNICEF Rwanda, 2021). There are very few modern tools that predict chronic stunting, and those that do in East Africa focus on acute malnutrition instead of stunting, or target national or county level rather than village level, where service delivery occurs in Rwanda, or are designed to analyse chronic stunting rather than predict it. (Asfaw et al., 2025; Jemil et al., 2026; Tadesse et al., 2025; Munyemana et al., 2024; Ndagijimana et al., 2026). As of now, there is no such mechanism in Rwanda to predict, detect, and deploy at the village level to end chronic stunting cases.

1.2 Problem statement

Although it has been 10 years since the introduction of policies promoting the use of CHWs, and around 45,000 CHWs have worked in Rwanda's 14,837 villages, the rate of childhood stunting persists at 27%, with CHWs working in the poorest villages and in rural areas having not necessarily influenced the condition of stunted children who are also in the poorest households (UNICEF Rwanda, 2021; NISR, MoH & ICF, 2025). This burden is persistent because not only is intervention capacity limited, but also intervention targeting – limited resources are being deployed reactively across all villages and not proactively where the risk is growing. Consequently, children may only be reached when interventions come too late, by about the age of two years.

Even current predictive machine learning systems have not remedied this situation. Munyemana et al. (2024) and Ndagijimana et al. (2026) are both Rwanda-focused studies that

utilized machine learning on DHS and household survey data. Nevertheless, they did not predict risk, but rather evaluated their programs after they had already taken place or classified them in a cross-sectional fashion. Children are already categorized as stunted from cross-sectional DHS data, with no operational lead time for interventions, in region-based studies (Asfaw et al., 2025; Jemil et al., 2026). The closest operational system, an AI system developed by USC, Microsoft AI for Good, and the Kenya Ministry of Health, predicts acute malnutrition at the county level in Kenya, instead of the outcome, granularity, and country for the Rwandan context in this case (Tadesse et al., 2025). There is no predictive system deployed at the village level for chronic stunting in Rwanda.

In addition, there is evidence that excessive targeting fails to have a positive impact, even when a large-scale nutrition intervention was initiated in Rwanda. Furthermore, even when there is this targeting, there is evidence that this was not enough to make an adequate impact on nutrition in Rwanda. A study testing the effectiveness of the USAID-funded Nutrition Program Gikuriro randomized a sample of interventions to document any improvement within 1 year for a set of core (non-indecomposable) child anthropometric outcomes: height-for-age, weight-for-age, dietary diversity, or anemia (McIntosh & Zeitlin, 2021). It highlights that programme implementation is weak in terms of guiding delivery to where and when interventions are most needed to bend the curve of stunting at the scale needed for Rwanda to reach its Vision 2050 commitments (Ministry of Health, 2022).

Kurinda is a multi-source early-warning machine learning tool to forecast village-level risk of chronic stunting three to six months ahead, which overcomes the identified gaps above through predictive, geographic prediction, and delivering actionable risk intelligence to CHWs and district nutrition officers via a web dashboard, a Kinyarwanda chatbot, and SMS alerts.

1.3 Project's main objective

To create a deployable machine learning early-warning system that predicts the likelihood of chronic childhood stunting (CCS) in a Rwanda village 3–6 months prior, by leveraging nutrition, agricultural, market, and satellite climate data to fill the current gaps in the quality and granularity of predictions and improve their accessibility for use by Community Health Workers and district nutrition officers to provide early preventive interventions before irreversible stunting sets in.

1.3.1 Specific Objectives

- To wrap up the literature about existing stunting prediction products and run an end-to-end exercise that involved collecting, compiling, and integrating data from various datasets, namely from the Concerned Health Information System (DHS), Rwanda Agriculture Board, eSoko, CHIRPS, and MODIS NDVI data sources, within the first three weeks of the project (Specific, Measurable, Achievable, Relevant, Time-bound).
- Designed, trained, and validated a LightGBM (gradient-boosted tree) model at the village level with SHAP-based explainability with a 3- to 6-month lead time that has a minimum AUC of 0.70 on held-out data; integrated the model into a three-channel delivery system (web dashboard, kinyarwanda chatbot, and SMS) within weeks four through eight.
- By week 9, to assess technical performance of the system (precision, recall, F1 score, AUC-ROC, lead-time accuracy) as well as operational impact (whether the proportion of villages achieving high-risk status was correctly identified before deterioration,

and/or whether the explanatory texts were of sufficient quality to be understandable in plain language, and accessible when offered in multilingual formats).

1.4 Research questions

1. Is it possible to generate operationally actionable forecasts of stunting risk at the Rwanda village level, with three to six months' lead time before clinical signs of stunting using data from multiple sources (nutrition, agriculture, markets, and satellite climate signals)?
2. How well does the predictive performance (AUC-ROC, precision, recall, F1) of a LightGBM model trained on fused multi-source data perform against a baseline model trained with only the DHS household indicators?
3. Does the explainability of models and the results of these models come to a level that allows operationalization with a simple message that can be easily understood and applied by the CHWs themselves?
4. Is there an evidence base showing that a three-channel delivery system (the web dashboard, the Kinyarwanda chatbot, and SMS alerts) contributes to extending the accessibility of the system to the most important (rural) part of the population using feature phones?

1.5 Project scope

- **Content focus:** The project will focus on predicting the risk of chronic childhood stunting (not acute malnutrition, wasting or overall food insecurity) at the village administrative level, and with a lead time of between 3 and 6 months. It targets two end-user groups, district nutrition officers and Community Health Workers, and enables the delivery of risk intelligence in three media: a web-based dashboard that is delivered using the Next.js framework; a chatbot in Kinyarwanda and English; and SMS using Africa's Talking platform.
- **Geographic scope:** Nationally available public data is used to apply the model to all 14,837 villages in Rwanda. The validation of the model will take place over the entire country during the operational deployment and field validation phase, but will be tested on a subset of 2-3 districts (13 districts were identified by the World Bank Stunting Prevention and Reduction Project as having the most severe stunting rates), for a period of nine weeks.
- **Time scope:** The entire Research, development, deployment, and validation cycle takes nine weeks (on a trimester basis) from 2026(05) till 2026 (37) with defense on 27/07/2026. The model is a 3~6 month prediction from the date of inference. The capstone does not include long-term monitoring, but it is considered future work.
- **Constraints:** There are several constraints recognized. First, the data acquired by DHS are obtained approximately every 5 years; supplementary signals from EICV (where available) and routine NECDP registration are used to address this. Second, the nine-week "window" only allows 2 or 3 districts to be qualitatively field-tested in the field, where the model-level deployment of 14,837 villages would require 2 years of field operational testing. Third, gradient-boosted tree models may have poorer MPP than deep learning approaches on the rich satellite imagery, but this is a compromise for the sake of explaining results to the frontline worker. Fourth, in

contrast to a diagnosis for a single child, the system is designed to predict the risk at a community level, a task that cannot be accomplished with a face-to-face clinical evaluation by a Community Health Worker. Fifth, reliance on open-source access to data feeds (eSoko, RAB, Google Earth Engine) must be maintained to enable prediction – source data feeds access would need to be reconfigured.

1.6 Significance and Justification

Kurinda will, for the first time, offer Rwanda's nutrition response system a targeting capability at the village level that goes beyond simply responding to damage to also target where risk is occurring, supplementing limited resources to help make a difference in villages that need it most.

The system supports targets for stunting reduction in Rwanda (2022) (Ministry of Health, 2022) and the United Nations Sustainable Development Goal (2.2) End all forms of malnutrition. Implementing cost-effective nutrition interventions could save 6.2 million children's lives around the world and add \$2.4 trillion in national economic benefits (World Bank, 2024); even a small speedup in Rwanda in geospatially aware stunting reduction is still valuable in terms of economic value.

The project represents an innovative, multi-source, geospatially aware, explainable machine learning approach to health in low-resource settings. It also supports the growing body of research on the implementation of AI in sub-Saharan Africa that features digital deployment using feature phones to those rural CHWs who do not use a smartphone and in their Kinyarwanda language.

1.7 Research Budget

This project is deliberately designed around free, open-source, and publicly available tools to ensure feasibility within a student capstone budget and to demonstrate that the solution is deployable in resource-constrained settings. The only essential variable cost is SMS credits for pilot testing of the alert channel.

Item	Description	Cost (USD)	Source / Notes
Python ML stack	pandas, scikit-learn, LightGBM, SHAP, GeoPandas	0	Open-source
Google Colab	Free GPU runtime for model training	0	Free tier
Google Earth Engine	CHIRPS rainfall, MODIS NDVI satellite data	0	Free academic access
GitHub	Source control, public repositories	0	Free for students
Next.js + Tailwind	Frontend dashboard framework	0	Open-source
FastAPI	Backend API framework	0	Open-source

Render hosting	Frontend + backend deployment (free tier)	0	Free tier
Google Gemini API	Chatbot LLM, RAG pipeline	0	Free tier
Africa's Talking SMS credits	Pilot SMS testing (~500 messages)	10	Required
DHS dataset registration	Public dataset access	0	Free
Internet & data bundles	Project communication, model training uploads	30	Personal
Total		40	

Table 1: Research Budget

1.8 Research Timeline

The nine-week implementation plan below maps deliverables to calendar weeks between May and July 2026, with weekly milestones and a strict priority ordering. If timeline pressure emerges, the chatbot feature is the designated cut item, as the core machine learning model and SMS alerts deliver the project's central scientific contribution.

Week	Dates	Focus	Key Deliverables
1	May 21–27	Setup & data acquisition	GitHub repos initialized, FastAPI + Next.js skeletons deployed to Render, DHS / CHIRPS / NDVI datasets downloaded
2	May 28 – Jun 3	Data pipeline & feature engineering	Village-month master dataset built, exploratory data analysis complete
3	Jun 4–10	Baseline model end-to-end	LightGBM baseline trained, exposed via /predict endpoint, Next.js fetching live predictions
4	Jun 11–17	Model improvement & explainability	Tuned model, SHAP explanations integrated, dashboard map functional
5	Jun 18–24	Dashboard polish	Officer view and CHW view functional, geospatial risk heatmap, village drill-down
6	Jun 25 – Jul 1	SMS alerts integration	Africa's Talking integration, Kinyarwanda alert templates, threshold triggering, two-way reply handling
7	Jul 2–8	Chatbot integration	Gemini + RAG over MoH/UNICEF nutrition documents, Kinyarwanda support, scoped query handling

8	Jul 9–15	Integration, testing, bug fixes	Full system tested end-to-end, deployed, demo-ready
9	Jul 16–22	Report, slides, rehearsal	Final report submitted, defense slides complete, three full rehearsals
—	Jul 23–26	Buffer	Final polish and rest
—	Jul 27	DEFENSE	Final defense and submission

Table 2: Project Plan (9-Week Implementation Plan)

CHAPTER TWO: LITERATURE REVIEW

2.1 Introduction

Looking into recent studies, this summary covers software tools plus machine learning work tied to child stunting and sudden malnutrition in poorer nations - especially places like East Africa and Rwanda. Most pieces examined come from scholarly articles or working system reports drafted since 2020, keeping things close to how decisions are made today. Material turned up via PubMed, PLOS One, Nature Scientific Reports, ScienceDirect, BMC series, along with unseen documents from UNICEF, the World Bank, the WFP, and Rwanda's health department. Papers got grouped around key ideas: first, school-built algorithms that sort out growth issues; second, live alert setups tracking hunger risks; third, what Rwanda itself has rolled out in public efforts; fourth, the backbone of data systems and fairness rules shaping real-world fixes. Around two dozen leads came up during search rounds. In the end, ten fit tightest - picked because they're fresh, line up well with methods used now, and connect clearly to the idea being explored.

2.2 Overview of Existing System

Some software tools aimed at tackling stunted growth and poor nutrition in kids fall into one of three kinds. One kind uses smart math systems built from household survey details, mostly from big health checks done over years. Take Jemil and team in 2026 - they used several number-crunching methods plus a way to show why predictions happen, testing on nearly seventy-six thousand young ones in East Africa using records between 2012 and 2022. Then there is Asfaw's work from 2025 - this time blending tree-style predictors with boost tricks based only on Ethiopia's numbers, hitting close to ninety-eight percent right answers when guessing different levels of slow growth. Another example comes from Phiri's research the same year - using layered learning codes on Malawi's home condition surveys to spot vulnerable children even before signs appear at clinics. In Rwanda alone, around four efforts stand out more than others. One step back, Munyemana and team in 2024 applied five different machine learning models to household surveys like DHS, MEIS, and LODA spanning 2015 to 2020 - but their aim sat firmly in reviewing past nutrition efforts, not predicting what comes next. Instead of looking ahead, Ndagijimana, Kabano, and colleagues in 2023 turned to machine learning to assess stunting right now in young kids across Rwanda, drawing from just one slice of DHS data. That thread continued when Ndagijimana's group in 2024 leaned into artificial neural networks, this time working strictly with the 2020 round of Rwanda's DHS. Just lately, by 2026, another shift: Ndagijimana teamed up with Nduwayezu to explore short stature in northern Rwandan children, folding climate and environment variables into their machine learning setup. Yet despite these moves, each of the four projects stays rooted in snapshots or backward glances at individual cases - no model reaches forward to forecast risks at the village level nor connects output to real-world rollout systems.

A system built around immediate warnings for food emergencies works like this. Since 1985, a program called FEWS NET - managed by USAID, an American government body - has been pulling together data from space-based sensors and field observations. Instead of waiting until disaster hits, it predicts sharp drops in food availability well ahead of time. Using an international ranking split into five levels, regions get grouped based on how bad conditions might become. Meanwhile, the World Food Programme's HungerMap LIVE uses algorithms to estimate current hunger rates in over ninety nations, pulling together survey

results, climate patterns, war zones, and economic trends. These tools aim their predictions at sudden shortages within large regions - cities, provinces - not long-term growth delays seen in small rural communities.

One step beyond sits a fresh batch of real-world AI uses - close to Kurinda's scale yet aimed elsewhere or placed in different regions. Out front stands an algorithm cooked up by USC Viterbi teamed with Microsoft's AI for Good Research Lab, alongside Amref Health Africa and Kenya's Ministry of Health (Tadesse et al., 2025). Instead of chasing broad impact, it pulls together medical records from over 17,000 clinics across Kenya. On top of that rides space-based views of crops and rain patterns. Together, they sketch out where young children may face severe hunger - not right now, but half a year ahead - and do so for each county. What shows up for officials is not raw code, but rather a working screen interface meant for health decision makers within Kenya.

2.3 Review of Related Work

What stands out first is how Jemil and team tackled childhood stunting using fresh machine learning methods across East Africa. Pulling data from nearly 77 thousand kids in DHS reports spanning a decade, their work leaned on several algorithm types to spot patterns. Instead of just listing factors, they brought in SHAP values along with rule-based mining to show what mattered most. Key drivers like household wealth gaps, mothers' schooling level, living outside cities, and spotty prenatal visits popped up clearly. Yet here's the catch - the whole setup looks backward, tagging kids who were already affected when surveyed. Even more telling, decisions came down to single children, not clusters such as villages or districts where Rwanda actually rolls out its health programs.

That study by Asfaw and team in 2025 pushed how we predict stunting using smarter methods. Machine learning ensembles were trained on data from Ethiopia's household surveys, focusing on different levels of stunting severity. Instead of just linking categories, they balanced skewed classes using SMOTE techniques. Accuracy jumped - up to 97.9 percent - with their chosen Random Forest approach sorting multiple severity stages well. Because results looked strong, the researchers stressed testing models live via apps on phones or websites. Frontline health staff could use such tools where help arrives late. A new setup called Kurinda takes that idea further, shaping it for Rwanda's villages and clinics.

One fresh look at child stunting in Rwanda comes from Munyemana and team, showing up in BMC Nutrition. Theirs is a first-of-its-kind analysis using machine learning tools tuned specifically to Rwandan conditions. Instead of linking arms across methods, they tested five models - SVM through to Decision Trees - on survey records spanning 2015 to 2020. Data flowed in from national health surveys plus two specialized sets: MEIS and LODA. What caught their eye was how current nutrition efforts lined up against slowed growth in infants below age two. Rather than pointing ahead, the approach digs into past numbers, asking what worked after the fact. It does not project future risks or shape alerts for next-phase actions. While it maps out traces of success in policy delivery, one space stays untouched - the shift toward real-time prediction. Evidence here sits still, rooted in description, not guidance.

Closest to Kurinda is what Tadesse and team outlined in 2025. Instead of jumping straight into solutions, they worked alongside Kenya's Ministry of Health to forecast child wasting half a year ahead. Data streams merged: routine clinic records through DHIS2 met space-based crop and weather measures. Results reached officials by way of a government

dashboard. Wasting was their focus - distinct from stunting - not just in cause but in timing and response route. Analysis happened per county, a broad scale compared to Rwanda's tighter village units. Kurinda borrows that blend - medical stats, farm conditions, satellite feeds - but shifts aim toward long-term growth failure. It zooms in deeper, down to small communities. Context changes too, fitting Rwanda's unique health setup.

Child growth issues in Rwanda's Northern Province take center stage in a 2026 analysis by Ndagijimana, Nduwayezu, Lind, and Mansourian - this work stands out as one of the few focused on local data. Climate patterns and natural surroundings shape their predictions about short stature in young children, blending these elements through machine learning tools. What sets their approach apart is how they weave together weather influences with landscape features to boost forecast accuracy. Yet Kurinda moves beyond this frame in several ways. One key contrast lies in reach: while the earlier effort zooms in only on the north, others cast a wider net across the entire country. Next comes its focus on sorting kids by present height compared to age, instead of predicting if they'll fall behind in growth within months. Then there's how it works per child only, skipping clusters like villages where health support actually reaches people. Another gap: missing pieces such as home surveys, local food costs, harvest numbers, and tools usable directly by community helpers. What Kurinda brings is an evolution - taking Ndagijimana et al.'s finding about weather patterns affecting stunting and pushing it ahead into a national setup that anticipates problems, groups results by locality, pulls varied inputs together, and can be rolled out practically across Rwanda.

Another layer shapes this effort, tied to past research. Not just UNICEF Rwanda in 2021, but their look at community health worker funding matters here - especially how Rwanda assigns three workers per village, a setup that fits neatly into what Kurinda built. Then there is the World Bank's 2024 plan on nutrition spending; it puts numbers behind smart early investments, showing why tools that guide where money goes make sense. That thinking feeds quietly into this work too.

2.3.1 Summary of Reviewed Literature

Author / System (Year)	Outcome Predicted	Geographic Granularity	Country / Region	Deployment Layer
Jemil et al. (2026)	Severe stunting (classification)	Individual child	East Africa (multi-country)	Academic only
Asfaw et al. (2025)	Stunting severity (multi-class)	Individual child	Ethiopia	Academic only
Munyemana et al. (2024)	Programme effect on stunting (retrospective)	Individual child	Rwanda	Academic only
Ndagijimana et al. (2026)	Stunting / height-for-age (classification)	Individual child	Rwanda (Northern Province)	Academic only

Phiri et al. (2025)	Childhood undernutrition risk	Individual child	Malawi	Academic only
FEWS NET	Acute food insecurity (IPC phases)	Country / sub-national	Global	Reports, dashboards
Tadesse et al. / USC-Microsoft-Kenya MoH (2025)	Acute child malnutrition	County	Kenya	Ministry dashboard
Kurinda (My proposal)	Chronic stunting risk (forecast)	Village	Rwanda	Dashboard + Chatbot + SMS

Table 3: Comparative Analysis of Existing Stunting Prediction Systems

2.4 Strengths and Weaknesses of the Existing System(s)

What stands out in current research is careful study design. Instead of just listing results, papers like Asfaw and team's 2025 analysis show how boosted decision trees spot kids likely to face growth delays; accuracy turns out solid when tested. Jemil's group, two years later, backs this up using combined models that sharpen predictions even more. Thanks to tools such as SHAP and LIME, newer reports dig into why certain factors matter, offering clues officials might act on. On the ground, warning setups including FEWS NET pull together weather data, market prices, and satellite views to track hunger risks across entire nations. Systems like HungerMap LIVE update constantly, showing real-time pressure points. A joint effort between USC, Microsoft, and Kenyan health authorities, managed by Tadesse in 2025, went further; it built working forecasts tied directly to government action for child nutrition.

Big problems show up in four ways. Not hitting the right mark comes first - most programs aim at sudden hunger crises or short-term lack of food, instead of long-standing growth delays in children, which stem from distinct causes, unfold slowly, and require separate fixes. Then there's scale trouble: the current tools work at national borders, regional zones, or district lines, but skip tiny villages where Rwanda's army of 45,000 local health helpers actually meet families. Another issue sits between research and reality: findings often stay locked in journals, never reaching those on the ground; even recent analysis by Asfaw and team points out that getting insights into the hands of caregivers remains undone. Not everyone has a smartphone. Many rural community health workers still rely on basic phones without internet. Most dashboards ignore this reality completely. Yet these workers make up a large part of frontline efforts. Kurinda's approach takes that into account from the start. Each limitation shapes how the system works. Design choices respond directly to real conditions on the ground.

2.5 General Comment and Conclusion

Looking into past studies shows that predicting child stunting with machine learning can actually be done. Combining different kinds of information like clinic records, farm conditions, satellite images, and food prices is where progress now pushes forward. Several researchers agree that the logical move ahead involves getting tools like these directly into the hands of local health staff. Yet none of the systems examined managed to cover all four

key aspects at once: focusing on long-term growth delays, working at the level of individual villages, built specifically for Rwanda, delivered through widely available communication channels. What sets Kurinda apart lies exactly here. By bringing together what others left separate, this effort adds something new instead of repeating earlier attempts.

CHAPTER THREE: SYSTEM ANALYSIS AND DESIGN

3.1 Introduction

This chapter describes the research design, system architecture, and modelling approach for the Kurinda early-warning system. The project follows a mixed-methods research strategy that combines quantitative supervised machine learning on multi-source structured data with qualitative validation of system usability through targeted interviews with Community Health Workers and district nutrition officers during the pilot phase. Software development follows an Agile-inspired iterative model with weekly increments aligned to the nine-week timeline presented in Chapter One. The design prioritises an end-to-end working pipeline at week three over feature completeness, with subsequent weeks devoted to model refinement and accessibility-layer development.

3.2 Research Design and Development Model

The research design integrates three components. First, a quantitative predictive modelling component fits machine learning models on a longitudinal village-month dataset constructed by spatially and temporally joining DHS household indicators, Rwanda Agriculture Board crop production records, eSoko market food prices, CHIRPS rainfall, and MODIS NDVI vegetation data, all referenced to NISR administrative shapefiles. Second, a qualitative validation component conducts semi-structured interviews with a purposive sample of CHWs and district nutrition officers in two to three pilot districts to assess perceived usability, explainability quality, and operational fit of the dashboard, chatbot, and SMS channels. Third, a comparative evaluation component benchmarks a multi-source fusion model against a DHS-only baseline to quantify the marginal predictive value of agricultural, market, and satellite signals.

The development model is iterative and incremental. Each of the nine project weeks closes with a working deployed system on Render, with one major capability added per week. The strategy is to ship a functioning ugly version of every layer before refining any layer, ensuring that even in adverse timeline scenarios, a complete defensible system is available.

3.3 Sampling Strategy

For the machine learning component, the unit of analysis is the village-month observation. With approximately 14,837 villages in Rwanda and a target longitudinal window of 60 months (2020–2025), the master dataset will comprise approximately 888,000 village-month rows after spatial join and temporal alignment, of which approximately 70% will be used for training, 15% for validation, and 15% for held-out testing. Stratified time-based splits will be used to prevent temporal leakage, ensuring that the test set strictly post-dates the training set.

For the qualitative usability component, purposive sampling will recruit between eight and twelve Community Health Workers and two to three district nutrition officers across the pilot districts. Sample size is set by the principle of thematic saturation given the bounded scope of usability validation and is consistent with capstone-scale qualitative work.

3.4 System Architecture

The Kurinda system is built as a modular, layered, end-to-end machine learning operations (MLOps) pipeline. Raw data is ingested from public sources, harmonized to a village-month grain, fed into a trained LightGBM model for inference, and surfaced through three distinct delivery channels. The system uses only free-tier or open-source components and is deployed on Render.

The architecture comprises seven layers. The data acquisition layer pulls from DHS (NISR registered access), RAB seasonal agricultural surveys, eSoko market price feeds, Google Earth Engine satellite collections (CHIRPS rainfall, MODIS NDVI), and NISR administrative shapefiles. The data integration layer performs spatial joins, temporal alignment, missing-data imputation, and feature engineering (lag features, rolling means, anomaly indices, cyclical seasonality encodings). Training happens using LightGBM trees, explained by SHAP, then saved. Prediction and explanation features appear through FastAPI routes alongside chat and message triggers. A web interface built with Next.js displays data while a local-language bot answers questions using health ministry documents. Messages reach users via an SMS platform connected to African telecom systems. All outcomes, actions taken, people involved, and system events go into a hosted PostgreSQL database. Both server and client applications run on Render, updated automatically whenever code changes land on GitHub.

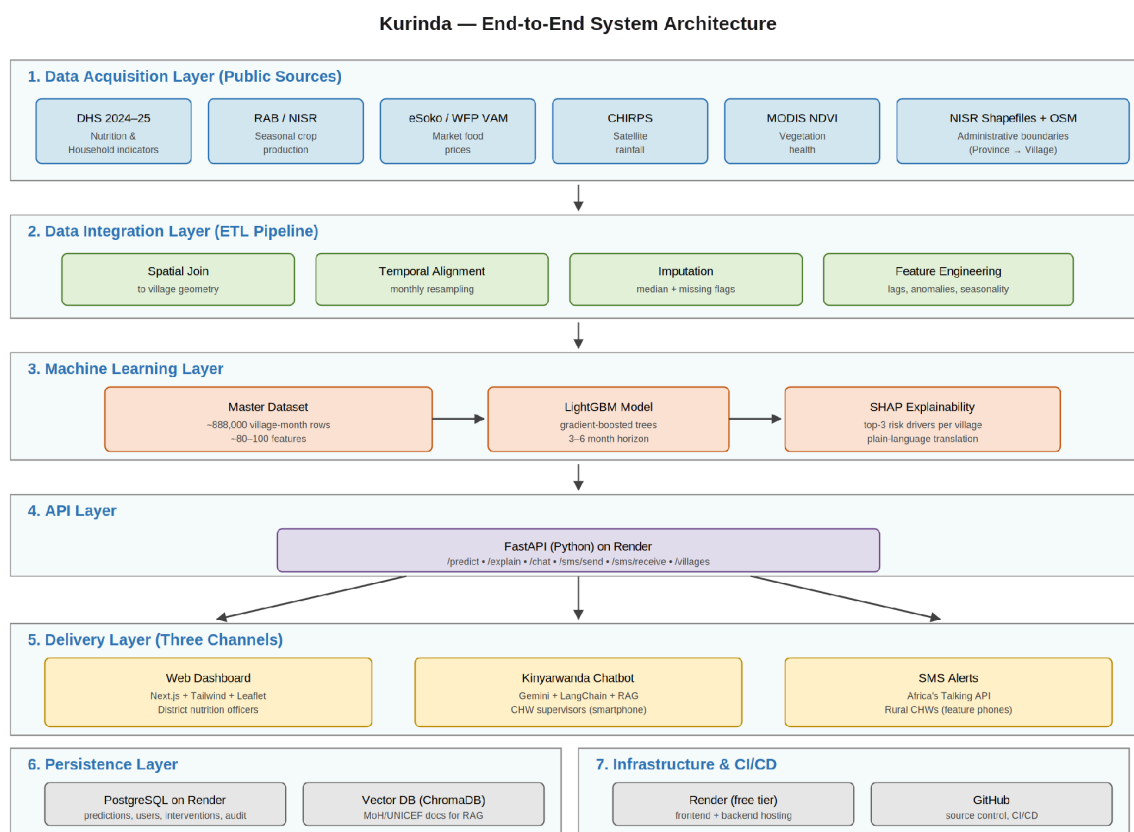


Figure 1: Kurinda End-to-End System Architecture

3.5 Data Definition and Acquisition

The project adopts a deliberately multi-source approach. Five distinct data sources are integrated to capture the full causal chain from climate shock to household nutrition outcome. The scale of the integrated dataset is approximately 888,000 village-month observations with roughly 80 to 100 engineered features. All sources are publicly accessible with no commercial licensing constraints; DHS access requires free registration through the DHS Program, and all other sources are open.

Category	Source	Representative Features
Nutrition & household	DHS 2024–25; EICV7	Height-for-age z-score, dietary diversity, exclusive breastfeeding rate, maternal education, antenatal care coverage, WASH indicators
Agricultural production	Rwanda Agriculture Board (RAB); NISR Seasonal Agricultural Survey	Seasonal yields (maize, beans, cassava, sweet potato), area cultivated, livestock ownership, crop diversity index
Market & food prices	eSoko Rwanda; WFP VAM; NISR CPI	Monthly retail staple prices, price volatility, terms of trade, and distance to the nearest market
Climate & satellite	CHIRPS (rainfall); MODIS NDVI (vegetation); ERA5 (temperature) via Google Earth Engine	Rainfall anomaly, standardized precipitation index, vegetation condition index, NDVI trend
Geospatial & structural	NISR administrative shapefiles; OpenStreetMap	Elevation, distance to health facility, distance to road, urban/rural classification, population density

Table 4: Key Predictive Input Features by Source Category

3.6 Machine Learning Training Pipeline

Starting off, the training process creates a consistent route from messy, mixed data to a working prediction tool. Five stages make up this flow. First comes gathering - data arrives through APIs, GeoTIFF files pulled down, or CSVs exported straight out. Then, checking and blending take place: structure gets verified, odd points are tossed, gaps are filled using either medians or smart guesses, while everything links together on a village-by-month level thanks to official map shapes from NISR. Next, shaping useful pieces happens - past snapshots appear at one, three-, and six-month marks, moving averages form alongside spread measures, unusual patterns stand out against old norms, time cycles get coded into signals that models can grasp. Fourth comes training plus checks. A LightGBM setup learns patterns using time-split data folds, while settings get fine-tuned through structured sweeps or smart search loops - calibration plots then show how well it guesses probabilities. Next up, clarity meets storage. Each forecast gets unpacked using SHAP scores; the strongest three influences become everyday sentences, clear like morning talk. The whole system locks down into a saved file, later pulled straight into an API when needed.

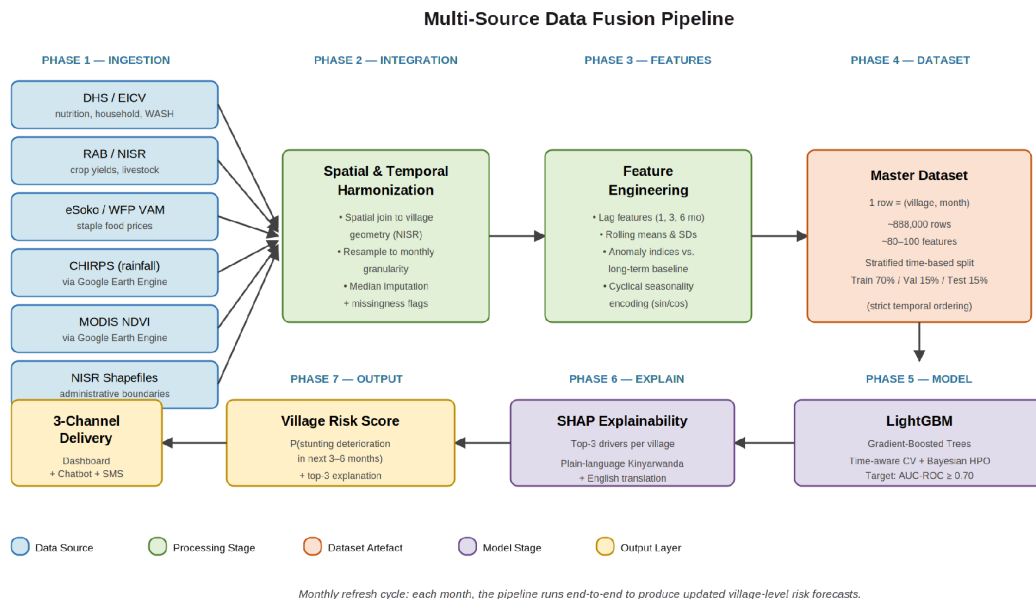


Figure 2: Multi-Source Data Fusion Pipeline

3.7 Model Comparison Architecture

Picking the best model means testing four options using identical data and time-based splits. LightGBM leads the pack - fast, handles categories and gaps well, works great on structured tables. XGBoost follows closely; it appeared in studies by Tadesse and Asfaw, both in 2025. Instead of stacking them together, we run Random Forest separately - it showed solid results on Ethiopia's DHS before, and gives us a clear reference for readable tree ensembles. A basic neural net joins in, just to see how older deep learning efforts like Phiri's from 2025 stack up. What tips the scale? Performance on unseen data is measured through ROC curves, how precise and complete predictions are at key thresholds, plus memory footprint, speed when scoring new cases, and whether SHAP fits smoothly into explanations. For conversation tasks, Google Gemini takes point as the main language engine. But another option opens beside it - one open-source pick, either Llama or Mistral - to test replies in Kinyarwanda side by side.

3.8 UML Diagrams

Out of four, each diagram maps part of how the system is built and acts. These visuals come together through Lucidchart or draw.io, sticking strictly to usual UML rules. Written details underneath give clear directions on what every image must show. Nothing gets left out when turning those explanations into proper diagrams.

3.8.1 Use Case Diagram

The key people involved are the District Nutrition Officer, CHW Supervisor, Community Health Worker, and System Administrator. Outside data comes from sources like DHS, RAB, eSoko, Google Earth Engine, and Africa's Talking. Some actions let users see a risk heatmap, while others allow zooming into village-level details. One path shows how SHAP explanations appear; another covers exporting an action report. Alerts arrive by SMS; visits get confirmed through replies. A chatbot handles system questions; interventions go on record when logged. The model updates when triggered; user accounts stay under control. Every scenario has clear starting conditions, step-by-step paths, possible variations, and outcomes noted in the diagram labels.

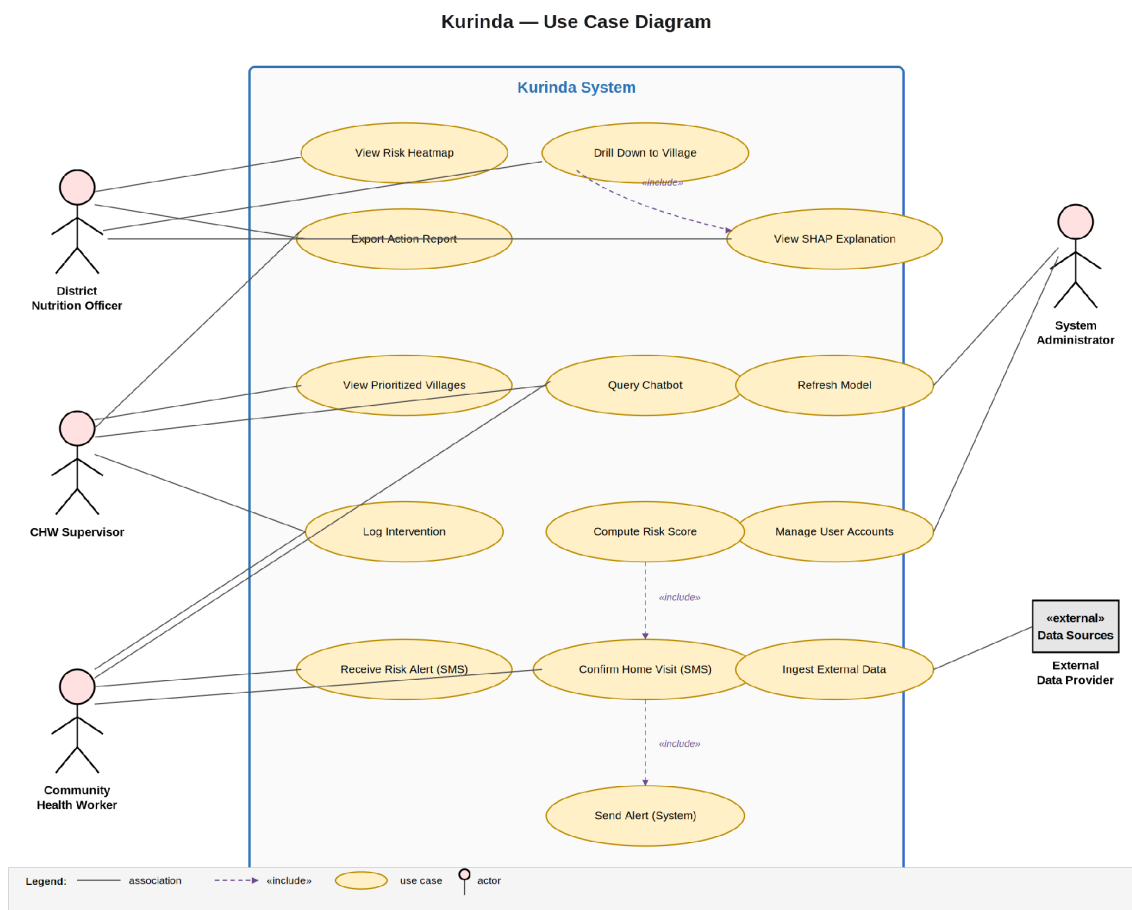


Figure 3: Use Case Diagram

3.8.2 Class Diagram

Village holds details like ID, name, area parts, position, and how many kids under five live there. Following that comes RiskPrediction - linked by village - with a score, time frame, when it was made, version of method used, plus its top three factors. Instead of just listing them, HouseholdIndicator ties into each settlement too, tracking home-level data. Then, the AgriculturalRecord shows farming activity across locations. On another path, MarketPriceRecord notes costs for goods over time. Weather facts land inside ClimateRecord, connected through location and date. People appear as User entries carrying identification, assigned duties, areas they cover, and phone numbers. These individuals may watch multiple villages, while each place could have several watchers - that bond lives outside either object. When danger levels rise, a RiskPrediction leads directly to an SMS being dispatched. Each alert travels via SMSAlert containing who got it, what it said, the delivery moment, and whether someone answered. Actions taken show up later as Intervention items tagged to teams or workers. Conversations happen separately within ChatSession records. Not every user handles alerts. Yet nearly all actions tie back to where things are.

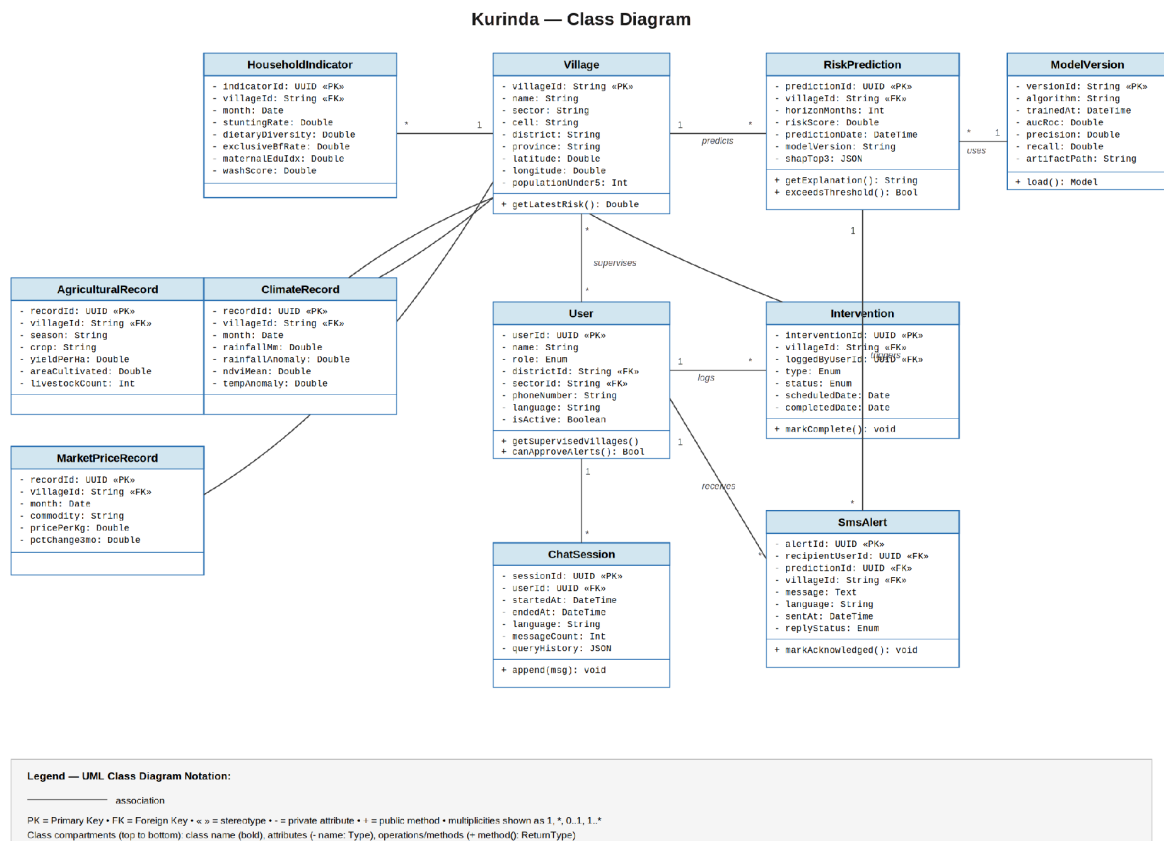


Figure 4: Class Diagram

3.8.3 Entity Relationship Diagram

A single village links to many risk predictions. Each prediction ties back to one specific location. Risk forecasts connect separately to multiple SMS alerts, though some might exist without notifications. One user can receive several messages over time. Predictions rely on data versions tracked through model version records. Households contribute indicators that feed into analysis. Farming details appear across agricultural reports tied to each place. Market prices shift alongside climate patterns stored in separate logs. Conversations between users happen within chat sessions logged system-wide.

Kurinda — Entity Relationship Diagram (ERD)

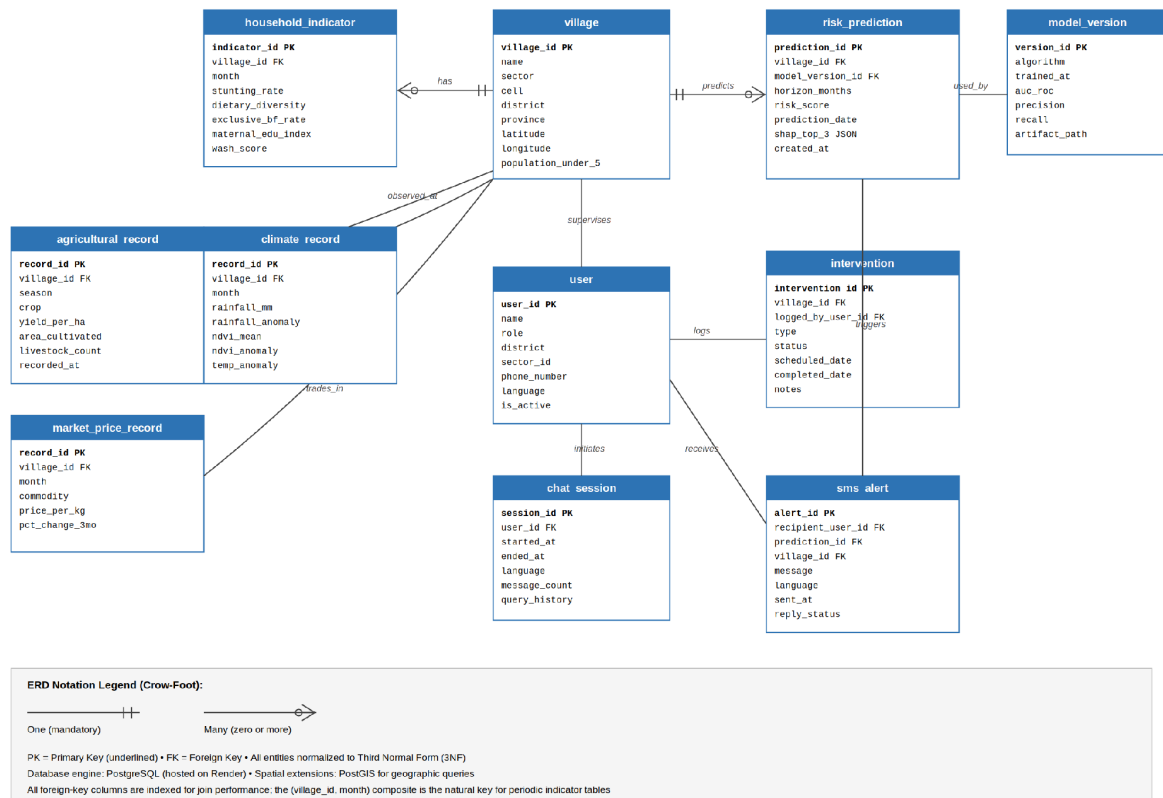


Figure 5: Entity Relationship Diagram (ERD)

3.8.4 Sequence Diagram (Risk Prediction and Alert Flow)

Step 1: A scheduled job in the FastAPI backend triggers a monthly model refresh. Step 2: Backend pulls the latest data from each source API into the PostgreSQL staging schema. Step 3: The ETL pipeline aggregates to village-month grain. Step 4: LightGBM model scores every village; SHAP top-3 drivers computed. Step 5: Risk scores are written to the risk_prediction table. Step 6: Alert trigger service queries new predictions, filters by threshold, and identifies affected CHWs. Step 7: SMS dispatched via Africa's Talking API in Kinyarwanda. Step 8: CHW receives SMS; replies '1' to confirm visit. Step 9: Webhook updates sms_alert.replyStatus. Step 10: Dashboard reads from risk_prediction to render updated heatmap.

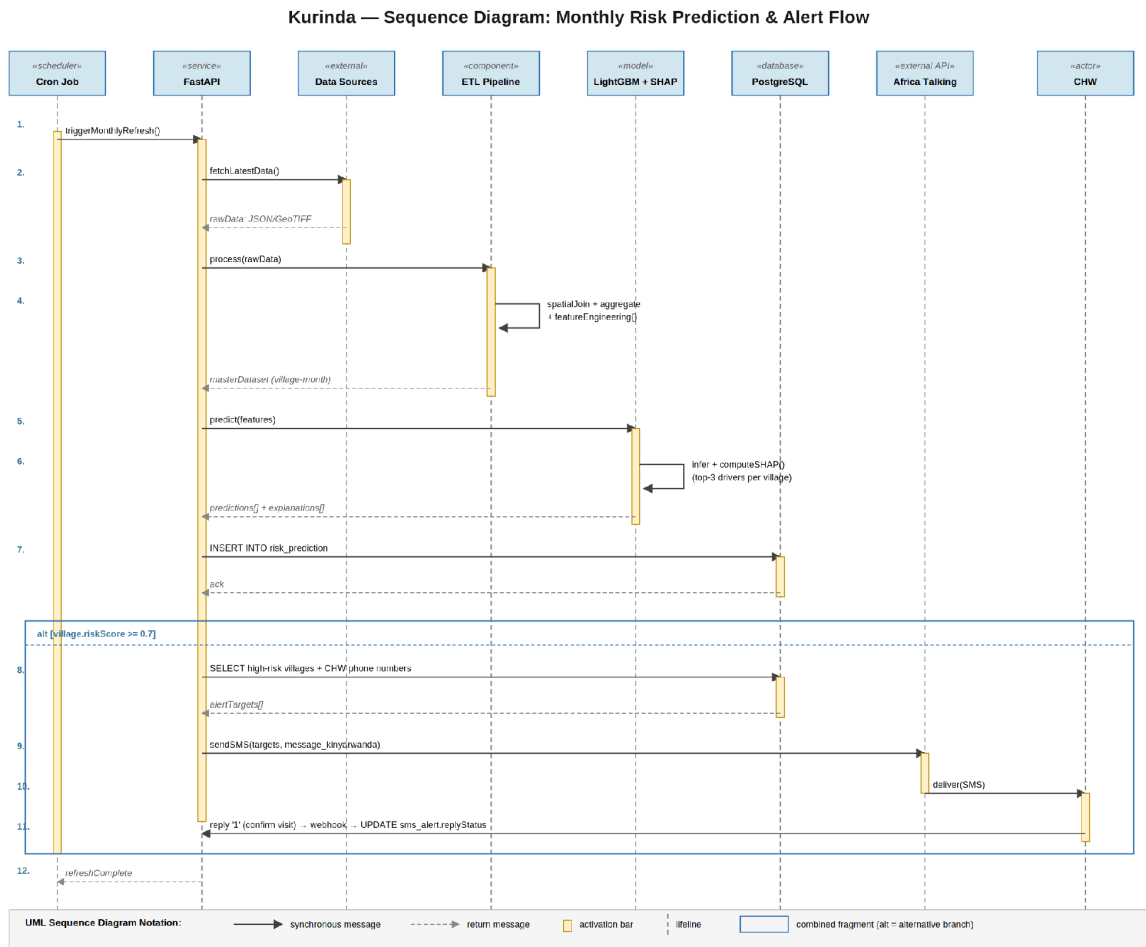


Figure 6: Sequence Diagram (Risk Prediction & Alert Flow)

3.9 Development Tools

Layer	Tool	Purpose
ML training	Python 3.11, pandas, NumPy, scikit-learn, LightGBM, XGBoost, SHAP, GeoPandas	Data wrangling, model training, explainability, geospatial processing
Notebook environment	Google Colab, Jupyter	Free GPU runtime, exploratory data analysis
Satellite data	Google Earth Engine	CHIRPS rainfall, MODIS NDVI access
Backend	FastAPI, Uvicorn	REST API serving model and SMS endpoints

Frontend	Next.js 15 (App Router), TypeScript, Tailwind CSS, shadcn/ui, Leaflet, react-leaflet, Recharts	Dashboard UI, geospatial visualization
Chatbot	Google Gemini API, LangChain, ChromaDB	LLM with retrieval-augmented generation
SMS service	Africa's Talking API	Kinyarwanda SMS alerts, two-way reply
Database	PostgreSQL	Predictions, users, interventions, audit
Hosting	Render (frontend + backend)	Unified deployment
Version control	Git, GitHub	Source code management, CI/CD
Documentation	Markdown, Overleaf (LaTeX)	Technical docs and final report

Table 5: Development Tools and Technologies

3.10 Ethical Considerations

Working within clear ethical rules shapes how this effort handles health data where resources are limited. To protect identity, the models learn only from broad summaries of public numbers, never touching personal details like names of kids or medical files. Access to detailed survey results comes through an official permit that allows study use alone. Making sure benefits spread widely drives another choice: answers come in Kinyarwanda, sent by basic text message so frontline workers without smartphones can still get insights others might miss. Instead of favoring high-end devices, it reaches those left out when systems depend on online screens. For one thing, clear reasoning matters. Each forecast comes with straightforward reasons behind it, drawn using SHAP methods, so people can question or challenge results instead of accepting them blindly. Starting differently now - clarity opens doors to trust. Not even close to replacing human insight, though the tool helps guide choices and sort cases faster, it never swaps out community health worker judgment. Right inside the display, it clearly states Kurinda suggests who to prioritize, not what's wrong. Another angle: working within existing structures counts. The rollout happens hand in hand with ALU teaching staff watching progress closely, plus sign-off from local health authorities when needed. Ending on storage rules, phone numbers sent via text stay saved just through the trial period. After that, everything gets wiped safely unless an official body says otherwise later.

References (APA Format)

All references follow the American Psychological Association (APA), 7th edition citation style. All sources are dated 2021 or later to ensure currency.

- Asfaw, F. A., Beyene, A. T., & Wassie, A. Y. (2025). Predicting stunting status among under-five children in Ethiopia using ensemble machine learning algorithms. *Scientific Reports*, 15, 8731. <https://doi.org/10.1038/s41598-025-03206-1>
- Jemil, H. W., Semayne, S. W., Kassaw, A. B., Gashu, K. D., & Otekunrin, O. A. (2026). Predicting severe stunting and its determinants among under-fives in Eastern African Countries: A machine learning algorithm approach. *PLOS ONE*, 21(1), e0340221. <https://doi.org/10.1371/journal.pone.0340221>
- McIntosh, C., & Zeitlin, A. (2021). Cash versus kind: Benchmarking a child nutrition program against unconditional cash transfers in Rwanda. *arXiv:2106.00213*. <https://doi.org/10.48550/arXiv.2106.00213>
- Ministry of Health, Republic of Rwanda. (2022). National Food and Nutrition Policy 2022–2027. Kigali: Government of Rwanda. https://www.moh.gov.rw/fileadmin/user_upload/policies/National_Food_and_Nutrition_Policy_.pdf
- Munyemana, J., Kabano, I. H., Uzayisenga, B., Cyamweshi, A. R., Ndagijimana, E., & Kubana, E. (2024). The role of national nutrition programs in stunting reduction in Rwanda using machine learning classifiers: a retrospective study. *BMC Nutrition*, 10(1), 98. <https://doi.org/10.1186/s40795-024-00903-4>
- Ndagijimana, A., Nduwayezu, G., Lind, T., & Mansourian, A. (2026). Machine learning techniques to model child low height-for-age in the Northern Province of Rwanda: The role of climatological and environmental factors and their interactions. *Clinical Epidemiology and Global Health*, 37, 102284.
- Ndagijimana, S., Kabano, I. H., Masabo, E., & Ntaganda, J. M. (2023). Prediction of stunting among under-5 children in Rwanda using machine learning techniques. *Journal of Preventive Medicine and Public Health*, 56(1), 41–49.
- Ndagijimana, S., Kabano, I. H., Masabo, E., & Ntaganda, J. M. (2024). Predicting stunting in Rwanda using artificial neural networks: A demographic health survey 2020 analysis. *F1000Research*, 13, 128.
- National Institute of Statistics of Rwanda (NISR), Ministry of Health (MoH) [Rwanda], & ICF. (2025). Rwanda Demographic and Health Survey 2024–25: Key Indicators Report. Kigali, Rwanda, and Rockville, Maryland, USA: NISR, MoH, and ICF.
- Tadesse, G. A., Ferguson, L., Robinson, C., Kuria, S., Wanyonyi, H., Murage, S., Mburu, S., Dodhia, R., Lavista Ferres, J. M., & Dilkina, B. (2025). Forecasting acute childhood malnutrition in Kenya using machine learning and diverse sets of indicators. *PLOS ONE*, 20(5), e0322959. <https://doi.org/10.1371/journal.pone.0322959>

UNICEF Rwanda. (2021). Community Health Program Investment Case in Rwanda. United Nations Children's Fund.
<https://www.unicef.org/rwanda/reports/community-health-program-investment-case-rwanda>

UNICEF, World Health Organization, & World Bank Group. (2024). Levels and trends in child malnutrition: UNICEF/WHO/World Bank Group joint child malnutrition estimates — 2024 edition. New York: UNICEF.

World Bank Group. (2024). Investment framework for nutrition 2024: Cost-effective interventions to reduce child mortality, stunting, and economic loss. Washington, DC: The World Bank. <https://www.worldbank.org/en/topic/nutrition/overview>